

An Earthquake Detection and Location Architecture for Continuous Seismograms: Phase Picking, Association, Location, and Matched Filter (PALM)

Yijian Zhou¹, Han Yue^{2,3}, Lihua Fang^{4,5}, Shiyong Zhou^{*2,3}, Li Zhao^{2,3}, and Abhijit Ghosh¹

Abstract

We developed an earthquake detection and location architecture for continuous seismograms that incorporates phase picking, phase association, location, and matched-filter techniques (PALM). The PALM architecture incorporates two modules: (1) PAL, the initial detection following picking, association, and location processes, and (2) match, expand, shift, and stack (MESS), a matched-filter detector that augments the template catalog. The effectiveness of PALM is demonstrated in building an early aftershock catalog for the 2019 Ridgecrest, California, earthquake. By comparing with Southern California Seismic Network arrival times, we show that the PAL picker combines the strengths of short-term average/long-term average and the kurtosis picker, realizing robust phase detection and precise picking. Our final MESS catalog is compared with two other matched-filter catalogs by Ross, Idini, *et al.* (2019) and Shelly (2020). We find that PALM directly recovers unbiased and detailed features in seismicity from continuous seismograms, which can be efficiently implemented to scan continuous waveforms without the need for visual inspection.

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[Supplemental Material](#)

Introduction

The detection and location of seismic events are basic procedures for generating earthquake catalogs. A catalog with high completeness and high resolution provides a detailed image of 3D fault geometry (Waldhauser and Ellsworth, 2002; Hayes *et al.*, 2012; Shelly *et al.*, 2016; Ross *et al.*, 2017), which facilitates the fault model construction to perform slip model inversions, (e.g., (Yue *et al.*, 2017; Sun *et al.*, 2018)). The spatiotemporal pattern and magnitude scaling also provide insights into the physical processes on fault, for example, slow-slip migration (Peng and Zhao, 2009; Kato *et al.*, 2016) and fluid diffusion (Di Luccio *et al.*, 2010; Chen *et al.*, 2012); The power-law distribution of events at different magnitudes is described by the *b*-value, which is observed to be correlated with stress level, fault roughness, and fault strength (Scholz, 1968; Schorlemmer and Wiemer, 2005; Spada *et al.*, 2013). In general, an ideal earthquake catalog should have high completeness, location precision, and consistent performance over a long time scale. Realizing this objective in modern networks, which typically include hundreds to thousands of stations, requires a high-performance detection and location

architecture to perform automatic and systematic scanning over continuous waveforms.

The detection of earthquakes is usually based on a network of seismic stations. Typical detection workflows involve (1) phase picking on single stations and (2) associating picks into events. Before the application of artificial intelligence (AI) seismology, automatic phase picking was usually realized with energy-based characteristic functions, for example, short-term-average over long-term-average (STA/LTA; Allen, 1978), which captures the abrupt change of amplitude in the seismograms. This type of methods presents high computational efficiency, but it tends to introduce a high false detection rate and

1. Department of Earth Sciences, University of California, Riverside, California, U.S.A., <https://orcid.org/0000-0002-7205-1769> (YZ); <https://orcid.org/0000-0002-0557-2839> (AG); 2. Institute of Theoretical and Applied Geophysics, Peking University, Beijing, China, <https://orcid.org/0000-0002-0950-6863> (LZ); 3. Hebei Hongshan Geophysical National Observation and Research Station, Peking University, Beijing, China; 4. Institute of Geophysics, China Earthquake Administration, Beijing, China; 5. Key Laboratory of Earthquake Source Physics, China Earthquake Administration, Beijing, China

*Corresponding author: zsy@pku.edu.cn

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does not discriminate P and S wave well. To overcome this limitation, proper association algorithms enable more accurate detection, though station records with a low signal-to-noise ratio (SNR) may still be missed. The phase association is usually realized by the temporal and spatial correlation between picks (Johnson *et al.*, 1997; Patton *et al.*, 2016; White *et al.*, 2019), but differences exist in the format of input picks. For example, Chen and Holland (2016) do not assign P or S to the picks and let the associator decide each pick to be P/S ; Zhang *et al.* (2019) need P and S picks, but they do not come in pairs, which leads to low efficiency.

In more recent years, two types of algorithms have presented better detection performance: AI and matched-filter technique (MFT). AI-based phase pickers incorporate the features of complete waveform (i.e., P and S phases), instead of solely the phase arrivals (Zhu and Beroza, 2018; Zhou *et al.*, 2019; Mousavi *et al.*, 2020). With numerous labeled training samples, AI models are able to detect and pick emergent phase arrivals. However, training samples are not always available, and pretrained models cannot guarantee performance comparable to that on the training set (Zhu *et al.*, 2019; Chai *et al.*, 2020). Such attempts in this direction include transfer learning (Zhu *et al.*, 2019; Chai *et al.*, 2020) and building a global training set (Mousavi *et al.*, 2019; Liu *et al.*, 2020). Even so, AI applicants still suffer from the difficulty in collecting and preprocessing training data, as well as building efficient training pipelines for big data. Thus, though a promising and fast-developing direction, AI pickers need additional case studies to validate their effectiveness and robustness.

Matched-filter augments the initial detections (i.e., templates) by cross correlating with raw data (Turin, 1960; Gibbons and Ringdal, 2006). This correlation-based algorithm can detect signals lower than the noise level and can thus obtain the most complete catalog (Shelly *et al.*, 2007). However, the detectability of MFT is determined by the diversity of template waveforms, which requires a rather complete catalog and high seismicity rate. Moreover, the cross correlation is computationally expensive, even with adopting a graphics processing unit (GPU) acceleration. These shortcomings limit the applicability of MFT to certain situations, for example, aftershock detection in which enough templates are available and only a few days of scanning is required. Thus, an ideal automatic detection workflow should provide reliable baseline detections as templates and an efficient MFT implementation based on GPU acceleration.

The location of earthquakes is usually based on arrival times picked for each event. Typical location processes include (1) absolute location, in which each event is located individually, thus minimizing the difference between observed and predicted travel times (Geiger, 1912; Klein, 2002), and (2) relative location, in which differential travel times for event pairs are measured to constrain the relative location (Waldhauser and Ellsworth, 2000; Zhou *et al.*, 2001; Trugman and Shearer, 2017). In such a workflow, the absolute location process obtains an accurate overall

location, and the relative relocation helps to resolve a detailed image of seismicity. An accurate absolute location calls for high-quality phase picking, proper choice of velocity model, adequate coverage of network, and fine-tuned location parameters (Husen and Hardebeck, 2010; Lomax, 2020). The picking of S arrival is especially important in this process because the S - P time constrains the length of the ray path, which reduces the trade-off between the origin time and epicentral depth in the inversion. However, S arrival is more difficult to precisely pick in comparison with P arrivals because its initial arrival is often contaminated by the tail of the P wave. This results in large uncertainty in determining the hypocentral depths. The relative relocation process resolves fine structure with the double-difference algorithm. Differential travel times measured by cross-correlation (CC) reach a theoretical resolution below the sampling rate (Frémont and Malone, 1987); thus, it has always been applied in company with MFT applications in recent years (Shelly *et al.*, 2016; Ross, Trugman, *et al.*, 2019), which results in both high detectability and high-resolution relocation.

The aforementioned progress makes possible a high-performance earthquake detection and location applied to modern networks. However, people still need to combine modules from different research studies, which possibly brings about low-efficiency pipelines and even errors. We acknowledge some successful cases following such a strategy (e.g., Mendoza *et al.*, 2019; White *et al.*, 2019), but a seamless and open-source architecture sequentially performing event detection, location, and match filter from raw data is still an urgent need of the seismic community. In this study, we present PALM, a newly developed detection and location architecture applied to raw continuous waveforms that incorporates phase picking, phase association, location, and matched filter. PALM is independent of any a priori information about earthquakes, which gives high-completeness detection and high-resolution location. We apply it to the 2019 M_w 7.1 Ridgecrest, California, (RC) aftershock sequence and compare our result with two other MFT catalogs: Ross, Idini, *et al.* (2019) and Shelly (2020) to validate its performance.

Method and Data

Workflow of PALM

The workflow of PALM includes two modules (Fig. 1): (1) PAL, that is, picking, association, and location, an STA/LTA-based algorithm for initial detection, and (2) MESS, that is, match, expand, shift, and stacking, a matched-filter detector that augments the initial PAL catalog. Each module contains detection and location stages and processes raw continuous waveforms. Details of PAL and MESS are presented in the following sections. Suggestions for software usage can be found in the supplemental material, available to this article.

PAL

The initial catalog is obtained by the PAL process, which includes three steps (Fig. 1): (1) picking: the picker first picks

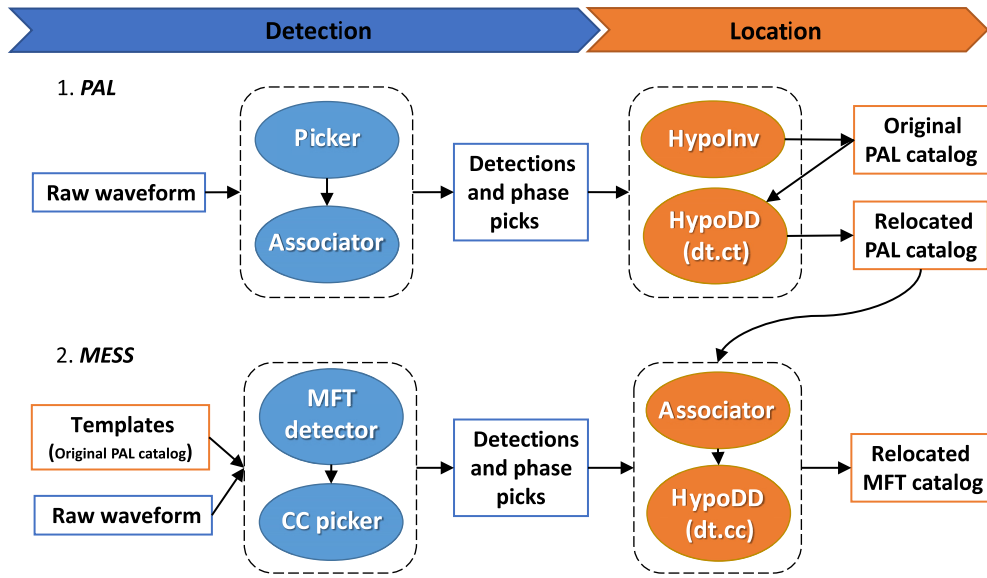


Figure 1. Workflow of phase picking, phase association, location, and matched-filter (PALM) techniques. Input and output results are denoted by squares. Detection and location operations are plotted in ellipse patches. Modules for detection and location purposes are plotted in blue and orange, respectively. MFT, matched-filter technique; MESS, match, expand, shift, and stack; PAL, picking, association, and location. The color version of this figure is available only in the electronic edition.

P and S arrivals from the continuous waveform; (2) association: the associator groups phases to different events for location purposes; and (3) location: events are located and relocated with standard algorithms, that is, HYPOINVERSE (Klein, 2002) and HypoDD (Waldhauser, 2001).

PAL picker. The PAL picker picks P and S arrivals in pairs, following a “detect and pick” strategy (Fig. 2). It is realized with a combination of STA/LTA (Allen, 1978) and the Kurtosis (Baillard *et al.*, 2013) algorithm, the definitions of which are

$$S_d(t_i) = \frac{\Delta t_{LTA}}{\Delta t_{STA}} \times \frac{\sum_{t_i}^{t_i + \Delta t_{STA}} d(\tau_i)}{\sum_{t_i - \Delta t_{LTA}}^{t_i} d(\tau_i)}, \quad (1)$$

in which S_d is the STA/LTA function for the input waveform (time series) d , Δt_{LTA} and Δt_{STA} are the window lengths for STA and LTA calculation, respectively. And

$$K_d(t_i) = \frac{\frac{1}{N} \sum_{t_i - \Delta t_K}^{t_i} (d(\tau_i) - \bar{d})^4}{\left[\frac{1}{N} \sum_{t_i - \Delta t_K}^{t_i} (d(\tau_i) - \bar{d})^2 \right]^2}, \quad (2)$$

in which K_d is the kurtosis function of data (d); Δt_K and N are the window length and number of sample points for kurtosis calculation, respectively; and d and $\bar{d}$ are the input data (time series) and mean value of $d(\tau)$ for $t_i - \Delta t_K < \tau_i \leq t_i$, respectively. The raw data are preprocessed by removing the mean

and linear trend, tapering, and band-pass filtering to the frequency band of local earthquakes. In the case of RC, we apply a 2–40 Hz band-pass filter. Then, the picking operation is performed on the processed data.

The P arrivals are detected on the Z-component with the STA/LTA algorithm (Fig. 2a). The characteristic function S_{p_det} is calculated on the kinematic energy of the vertical component:

$$S_{p_det}(t) = S_z^2(t), \quad (3)$$

in which z is the Z-channel velocity. The Δt_{STA} is relatively long for detection purposes. For RC data, we set $\Delta t_{STA} = 0.8$ s and $\Delta t_{LTA} = 6$ s. A detection is declared at the triggering time t_{p_trig} if S_{p_det} reaches the triggering threshold S_{p_thres} , that

is, $S_{p_det}(t_{p_trig}) = S_{p_thres}$. We set $S_{p_thres} = 12$ in RC based on the estimation of the SNR. Precise P arrivals (t_p) are picked around the triggering time by an STA/LTA operation calculated on the vertical energy, but with a shorter Δt_{STA} (Fig. 2b). We set $\Delta t_{STA} = 0.4$ s and $\Delta t_{LTA} = 2$ s for P -arrival picking in RC. The initial picking is made at the peak of the STA/LTA function, which is then corrected on the Z-velocity by finding the first peak backward.

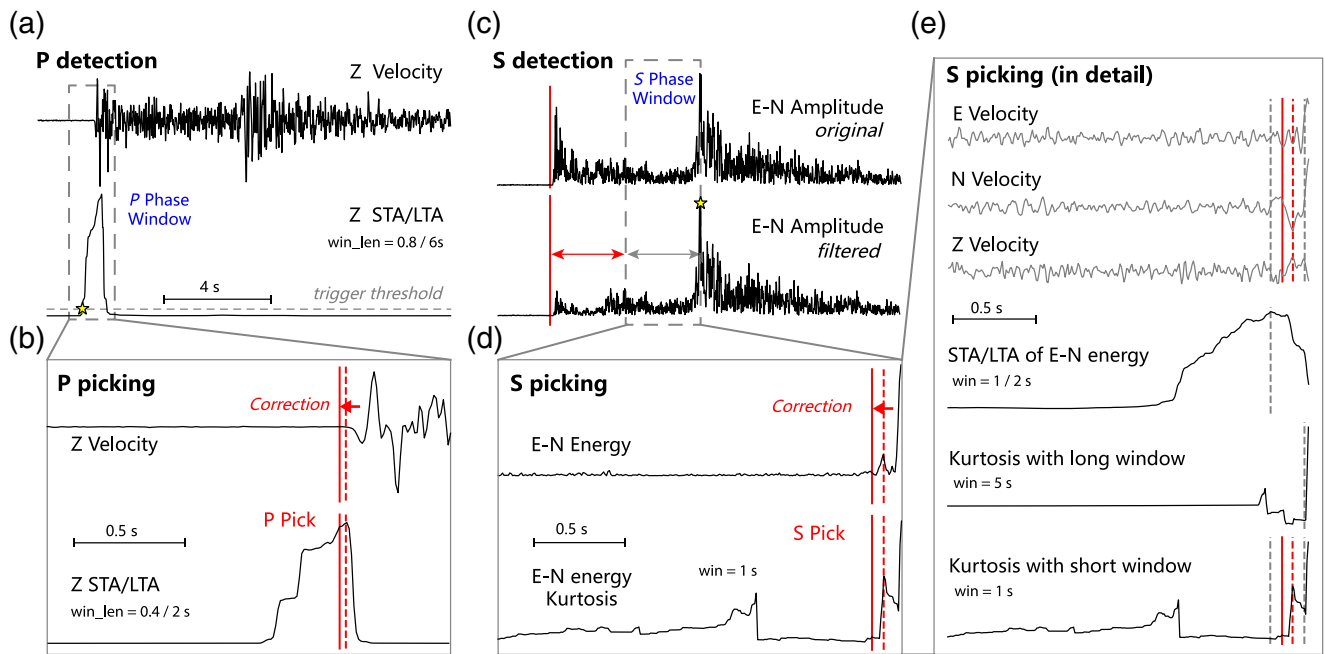
S -wave arrivals are detected on the horizontal components (E and N channels) (Fig. 2c). The triggering time for S -wave t_{s_trig} is defined as the time when the S -wave amplitude $A_s(t)$ reaches its peak value:

$$A_s(t_{s_trig}) = \max[A_s(t_i)], \quad \text{for } t_p < t_i < t_p + \Delta t_S, \quad (4)$$

in which Δt_S is the searching window for S wave following the picked P -arrival time t_p . The S -wave amplitude A_s is defined as

$$A_s(t_i) = \begin{cases} \sqrt{x^2(t_i) + y^2(t_i)} \times P_{\text{filter}}, & \text{for } t_p < t_i \leq t_p + \Delta t_{\text{filter}} \\ \sqrt{x^2(t_i) + y^2(t_i)}, & \text{for } t_p + \Delta t_{\text{filter}} < t_i < t_p + \Delta t_S \end{cases}, \quad (5)$$

in which P_{filter} is a polarization filter based on principal component analysis (see supplemental material) and Δt_{filter} is the time window in which the polarization filter is applied. We set $\Delta t_{\text{filter}} = 2$ s in RC to remove the possible contamination from



the P wave. The S arrival is thus constrained between $t_P + (t_{s_trig} - t_P)/2$ and t_{s_trig} (namely “ S window”).

Precise S -arrival time t_S is picked by the kurtosis of the horizontal energy, with a small Δt_K (i.e., K_{short}) to capture the emergent change in amplitude (Fig. 2d,e). The correct peak in K_{short} is further constrained by STA/LTA, S_{s_det} , and long-window kurtosis K_{long} in the S window (Fig. 2e):

$$\begin{cases} S_{s_det}(t_{min}) = \max(S_{s_det}) \\ K_{long}(t_{max} + \Delta t_{corr}) = \max(K_{long}) \end{cases}, \quad (6)$$

in which t_{min} and t_{max} are the earliest and latest boundaries of the peak in K_{short} and Δt_{corr} is the time correction made by finding the first trough backward. In RC, we set $\Delta t_{STA} = 1$ s and $\Delta t_{LTA} = 2$ s to calculate S_{s_det} and set $\Delta t_K = 1$ s and $\Delta t_K = 5$ s to calculate K_{short} and K_{long} , respectively. The initial picking is made at the peak of K_{short} , which is then corrected on the horizontal energy by finding the first peak backward.

The PAL picker provides precise P and S picks for correctly detected phases. However, it may yield false detections due to impulsive noise signal (Withers *et al.*, 1998; Yue *et al.*, 2018). We conduct a subsequent phase association process to remove false detections.

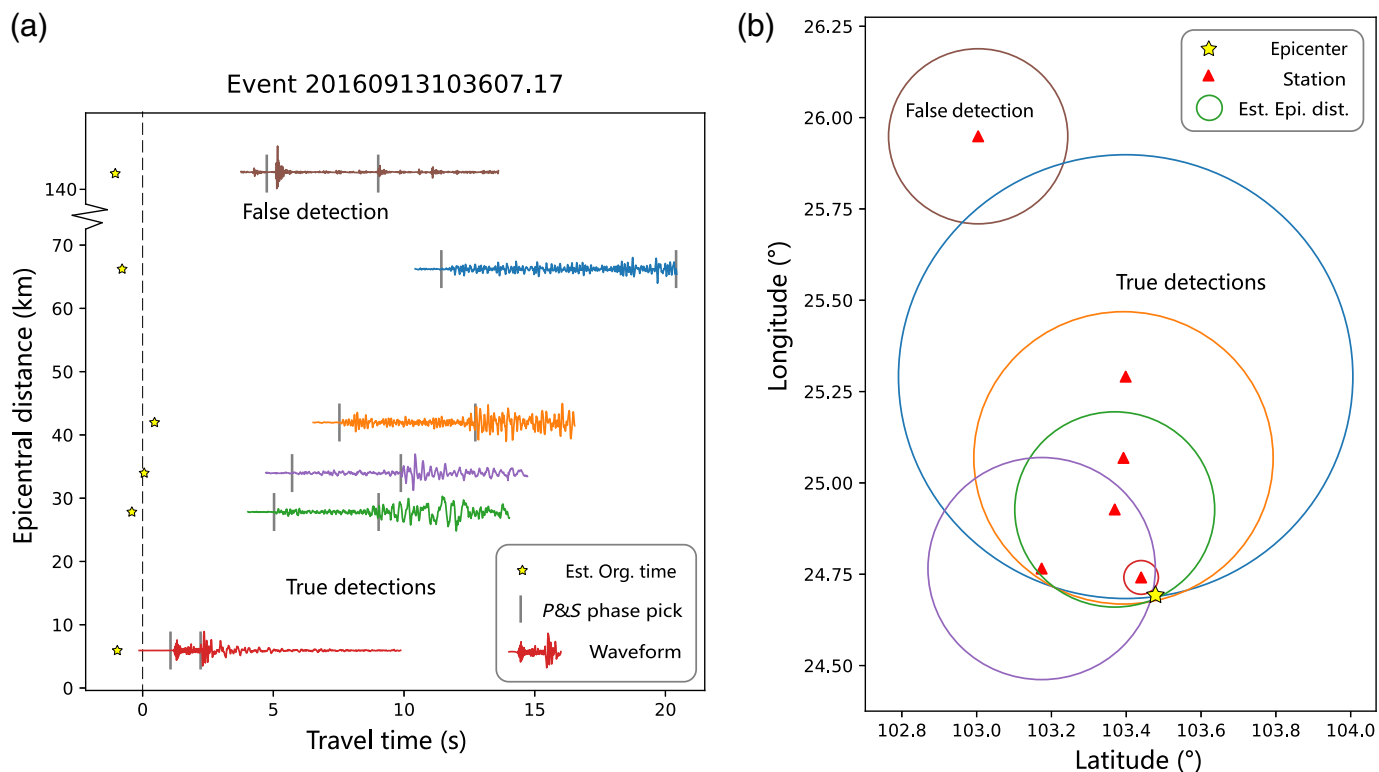
PAL associator. The P/S arrivals are associated with different events based on their spatial and temporal relations at different stations (Fig. 3). The temporal association is realized by clustering the estimated origin times (Fig. 3a). The origin time t_o at a given station is estimated as

$$t_o = \frac{t_P \times V_P - t_S \times V_S}{V_P - V_S}, \quad (7)$$

Figure 2. Algorithm of PAL picker. (a) Detection of P arrival. Z-channel waveform (velocity record) and short-term average/long-term average (STA/LTA) function are plotted in black lines. The trigger level and time are plotted by the dashed horizontal gray line and yellow star, respectively. The P phase window is bounded by gray-dashed rectangles. (b) Picking of P wave. Red dashed and solid lines plot initial and corrected (final) P picks, respectively. (c) Detection of S arrival. Original and principal component analysis (PCA)-filtered horizontal (E-N) channel amplitude are plotted in the top and bottom panels, respectively. The time of maximum filtered amplitude is marked by the yellow star. The S -phase window is marked as the gray-dashed rectangle and gray double arrow. The red double arrow marks the first half of the window between P and the yellow star (i.e., P window). (d) Picking of S wave. The labels are identical as in panel (b). (e) Detailed illustration of the S -picking algorithm. Characteristic functions are plotted in black lines. The earliest and latest boundaries derived from STA/LTA and the long-window kurtosis picker are marked by gray-dashed lines. The initial and final S picks are marked by red-dashed and solid lines, respectively. The color version of this figure is available only in the electronic edition.

in which V_P and V_S are the average P - and S -wave velocities in the crust, respectively. In the case of RC, phase picks are grouped into the same events if their origin-time estimations deviate within 2 s. Temporally associated phase picks are retained for further processing in the spatial association.

Epical distances can be roughly estimated from $t_S - t_P$, drawing a location circle centered by the station (Fig. 3b). For point source (earthquake), these circles from different stations converge near the real epicenter. This convergence serves as a criterion for spatial association, which can eliminate arrivals inconsistent with the common convergence point (i.e., false



detection). This spatial association is realized by a 3D grid search for the epicentral location that minimizes the travel-time residual ε defined as

$$\varepsilon = \sum_{i=1}^N \varepsilon_i = \sum_{i=1}^N |(t_o + T_{P,i}) - t_{P,i}|, \quad \text{for } \varepsilon_i < \varepsilon_{\max}, \quad (8)$$

in which $T_{P,i}$ and ε_i are the respective P -wave theoretical travel time and residual at the i th station, respectively, and N is the number of stations with travel-time residuals that fall below the threshold $\varepsilon_{\max}$. We set $\varepsilon_{\max} = 1.5$ s for the RC events, considering the uncertainty in travel-time estimation. A minimum number of stations is required for a robust location, which is set as $N \geq 4$ here.

Finally, we estimate the local magnitude M_L using the S -wave amplitude, according to the classical definition by Richter (1935):

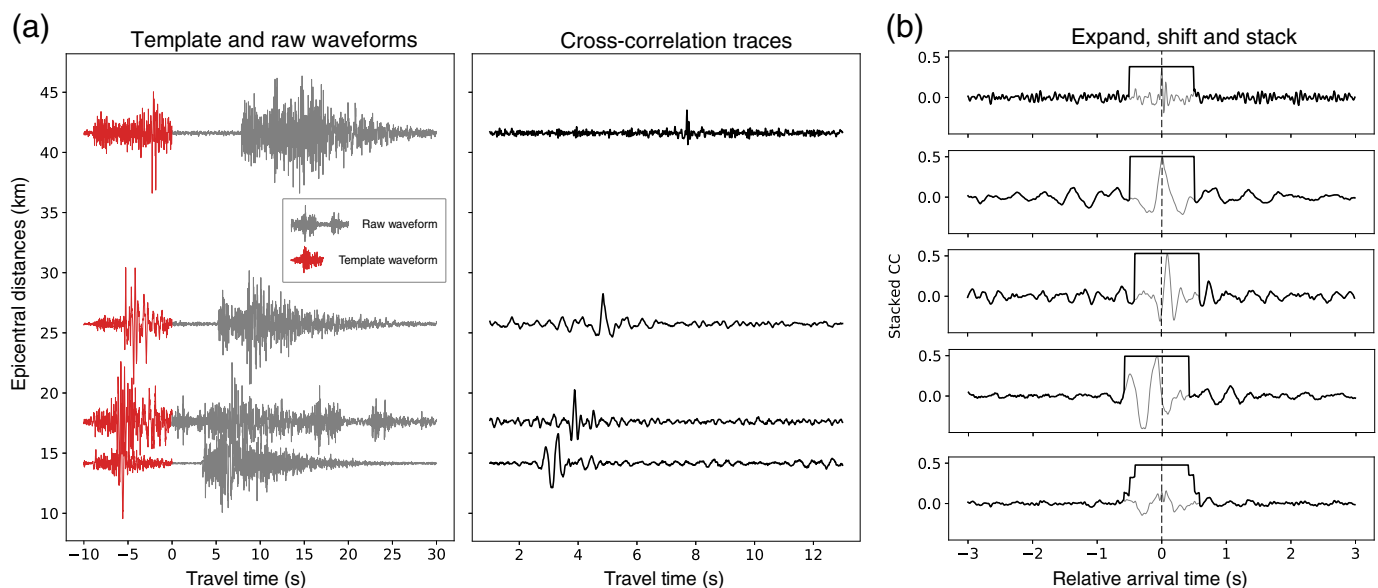
$$M_L = \text{median}(\log_{10}(A_{S,i}) + \log_{10}(r_i) + 1), \quad (9)$$

in which $A_{S,i}$ is the S -wave amplitude (in μm) at the i th station and r_i is the corresponding hypocentral distance in kilometers. We do not include distance attenuation or the station correction term for wider applicability because they are path and site dependent and are not available in most cases (Boore, 1989). This results in a similar magnitude scale in California and thus is suitable for the case of RC (Hutton and Boore, 1987; Uhrhammer *et al.*, 2011).

Figure 3. Phase association process. (a) Temporal association. E-channel waveforms at different stations are plotted in different colors. Vertical solid-gray lines mark the picked P - and S -arrival times. Yellow stars denote the estimated origin times. (b) Spatial association. Red triangles show the locations of seismic stations. Circles are centered at the stations with corresponding epicentral distances estimated from the picked P - and S -arrival times. Yellow stars denote the estimated epicenter. The color version of this figure is available only in the electronic edition.

The PAL associator utilizes pairs of P - S picks to realize stable and efficient associations, which is different from previous algorithms (e.g., Chen and Holland, 2016). This may result in more missed detections when the network is rather sparse or the noise level is high. This problem can be solved with the following MESS detection procedure.

PAL locator. We adopt standard location and relocation procedures in the location stage because the PAL detector and associator produce absolute arrival times. The detected events are first located using HYPOINVERSE (Klein, 2002), which is an absolute location method based on gradient descent inversion. Then, the locations are refined by HypoDD (Waldhauser, 2001), using the catalog-derived differential time (namely $dt.ct$) (Fig. 1). The original PAL catalog serves as template events for MESS detection, and the relocated PAL location is applied in relocating MESS detection results.



For RC events, we employ the 1D velocity model used by Shelly (2020), which is a 1D approximation of the 3D Community Velocity Model for Southern California model (Kohler *et al.*, 2003) (Fig. S9). In the absolute location using HYPOINVERSE, we apply a cosine distance weighting factor that changes from 1 at 30 km to 0 at 90 km and a residual weighting function that ranges from 1 at 0.3 s to 0 at 0.9 s of maximum residual for *P* and *S* picks. In the HypoDD relocation, we set the parameter WDCT to 5 km and apply four iterations of inversion, considering the average interevent distance.

MESS

MESS detection. The initial PAL detections are augmented by MESS, a matched-filter detector that consists of four steps: match, expand, shift, and stack. In this process, both event detection and phase picks are realized by cross correlation, which is accelerated by the GPU. The details of MESS are described as follows:

1. Match: the similarity between template and raw data is quantified by the cross correlation (Fig. 4a):

$$CC_0(t_i) = \frac{\sum_{\tau_i=t_i}^{t_i+\Delta t_{\text{temp}}} d_{\text{raw}}(\tau_i) \times d_{\text{temp}}(\tau_i - t_i)}{\sqrt{\sum_{\tau_i=t_i}^{t_i+\Delta t_{\text{temp}}} d_{\text{raw}}^2(\tau_i)} \times \sqrt{\sum_{\tau_i=0}^{\Delta t_{\text{temp}}} d_{\text{temp}}^2(\tau_i)}}, \quad (10)$$

in which d_{raw} and d_{temp} are the raw and template waveforms, respectively, and Δt_{temp} is the length of the template waveform. For the RC data, the template uses 10 s windows, from 1 s before to 9 s after the *P* arrival, to cover both *P* and *S* waves for accurate detection. Both template and raw waveforms are preprocessed in the same way as in the PAL process.

Figure 4. Algorithm of MESS. (a) Matched filter. Template and raw waveforms are plotted in red and gray lines, respectively. The resulting cross correlations are plotted in black. (b) Stacking strategy. Original and expanded cross correlations are plotted in gray and black lines, respectively. The color version of this figure is available only in the electronic edition.

2. Expand: a peak value expanding operation is imposed on the cross-correlation functions (Fig. 4b):

$$CC_{\text{exp}}(t_i) = \begin{cases} CC_{\text{peak}}, & \text{for } t_{\text{peak}} - \Delta t_{\text{exp}}/2 < t_i \leq t_{\text{peak}} + \Delta t_{\text{exp}}/2, \\ CC_0(t_i), & \text{otherwise} \end{cases}, \quad (11)$$

in which CC_{peak} and t_{peak} are the amplitude and time of the CC_0 peaks, respectively, and Δt_{exp} is the length of expansion. The expansion operation is designed to reconcile location differences between the templates and detected events because a direct summation fails to align peak CC values at the origin time of the detected event, which limits the detectability (see the comparison in Fig. 4b). We set $\Delta t_{\text{exp}} = 1$ s in RC, which corresponds to a searching radius of about 3 km around the template event.

3. Shift: The expanded cross correlation is shifted by the *P*-wave travel time of the template event (Fig. 4b):

$$CC(t_i) = CC_{\text{exp}}(t_i + T_p), \quad (12)$$

in which T_p is the *P*-wave travel time of the template event. Given that the template event has a similar location to the detected event, this time shift can roughly align the cross correlations from different stations.

4. Stack: The migrated CCs from different stations are stacked (Fig. 4b):

$$CC_{\text{stack}}(t_i) = \frac{1}{N} \times \sum_{i=1}^N CC_i(t_i), \quad (13)$$

in which N is the number of stations. Multitrace stacking suppresses uncorrelated random noises, further reducing the false detection rate. We set the detection criterion as $CC_{\text{stack}} \geq 0.25$, and the origin time t_o is determined as the peak time on CC_{stack} .

The P - and S -arrival times are then picked by waveform cross correlation. The template window for CC picking ranges from 0.5 s-pre to 1.5 s-post the P arrival and from 0.5 s-pre to 2.5 s-post the S arrival, considering the length of the signal in RC. The S amplitude is also individually measured for magnitude estimation using the same method as that for PAL detections.

MESS relocation. The MESS detections are relocated by HypoDD (Waldhauser, 2001), using the correlation measured differential time as the dt_{cc} file (Fig. 1). Detections from different templates are first associated based on their origin time. In RC, we associate detections with estimated t_o within 2 s as one detection, and its origin time is the t_o estimation by the template with the maximum CC_{stack} value. The differential travel time between the template and detected event is obtained by

$$\Delta T = t_{cc} - t_o - T_{\text{temp}}, \quad (14)$$

in which t_{cc} is the correlation picked arrival time, t_o is the origin time of the detected event, and T_{temp} is the travel time of the template event. The initial location of the templates is shifted to the relocated PAL catalog location, whereas the initial location of newly detected events is set to the average of template locations that detected it.

For RC events, we adopt the same velocity model as for the PAL location. For robust relocation, each event is required to be detected by at least three templates with dt from four stations. We utilize only dt_{cc} to relocate MESS detections, with the parameter WDCC set to 4 km for five iterations of relocation.

Southern California Seismic Network (SCSN) data

We use the aftershock catalog of the 2019 Ridgecrest earthquake to validate the performance of the PALM pipeline. The detected arrival time and relocated catalog are compared with published results. We utilize publicly available SCSN data to build the RC aftershock catalog (Cochran *et al.*, 2020; Hauksson *et al.*, 2020). Broadband and short-period stations that distribute within approximately 100 km from the mainshock epicenter are incorporated, which covers the source region well. The average distance between stations is about 10–20 km (Fig. S1). This selection of stations is similar to that used by Ross, Idini, *et al.* (2019) and Shelly (2020). We apply

PALM from 4 July to 16 July 2019 enabling direct comparison with the Ross, Idini, *et al.* (2019) and Shelly (2020) catalogs.

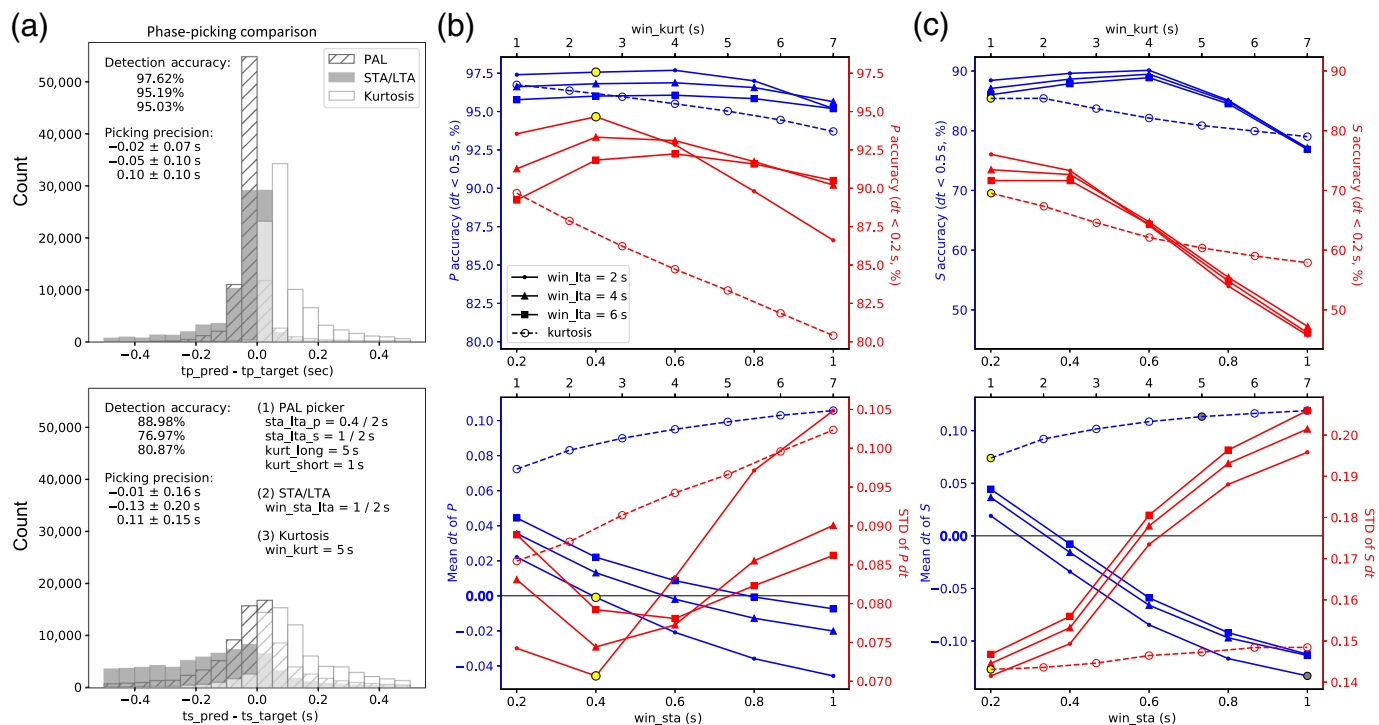
Results and Discussion

Phase-picking results

Phase picking is the fundamental step for event detection and location in the PAL architecture. We test the PAL picker on the PAL-recalled SCSN events and use SCSN picks as a reference, which contains about 100,000 P and S pick pairs. We made a detailed comparison between PAL, STA/LTA, and the kurtosis picker picking results. Here, both STA/LTA and the kurtosis picker pick the peak time in the characteristic functions, as defined by equations (1) and (2), respectively. The choice of channel is also kept the same with PAL picker: pick P arrivals on Z -vertical velocity waveforms and S arrivals on the east–north–horizontal energy. The picking operation is performed over a ± 1 s time-window reference to the SCSN arrival times.

We first compare the distribution of picking deviation (dt) for the three pickers (Fig. 5a). The phase-detection accuracy is defined as the ratio of picks with $dt < 0.5$ s to the total number, which measures the stability of a picker. The phase-picking precision is expressed as the mean $\pm$ standard deviation (st. dev.) of the detected phases, whereby the mean value evaluates the systematic picking deviation and the st. dev. evaluates the picking concentration. As shown in Figure 5a, the PAL picker outperforms STA/LTA and the kurtosis picker for both P and S picks, being stable in detection and precise in picking and having little systematic deviation. In contrast, STA/LTA tends to pick ahead of SCSN and has low picking precision. This bias is more serious for S waves, which has a low SNR. The kurtosis picker has high picking precision, but it results in lagged picks, which has also been reported by previous studies (Baillard *et al.*, 2013; Ross and Ben-Zion, 2014). Further improvement can be made with AI methods (Zhu and Beroza, 2018; Zhou *et al.*, 2019; Mousavi *et al.*, 2020), especially for the S -wave picking. However, as stated in the Introduction, the generalization of AI methods is still a problem and needs further investigation, but this is out of the scope of this article.

We further explore the performance of STA/LTA and the kurtosis picker with different window lengths (Fig. 5b,c), which illustrates the advantage in PAL picker because it combines the strengths of the two traditional pickers. In P picking (Fig. 5b), STA/LTA has higher detection ability and picking precision than kurtosis, under most combinations of window lengths. For the STA/LTA method, the mean dt drops with increasing Δt_{STA} , and the best performance is obtained with $\Delta t_{\text{STA}} = 0.4$ s and $\Delta t_{\text{LTA}} = 2$ s. When picking S arrivals (Fig. 5c), the kurtosis picker is more robust to different window lengths and has significantly higher picking precision compared with STA/LTA; however, the kurtosis picks are consistently later than the SCSN picks by about 0.1 s, which is caused by the emergent arrival of S wave. Though the S -wave initial arrival shows up as a peak in the kurtosis function, the growing amplitude



afterward leads to a larger kurtosis amplitude (e.g., Fig. 2 and Figs. S3–S7). On the other hand, STA/LTA gives consistent early picking when $\Delta t_{STA} > 0.5$ s. Thus, the STA/LTA and long-window kurtosis provide a stable constraint on S-arrival times. Based on this constraint, the PAL picker utilizes short-window kurtosis to reveal a more emergent change in amplitude, without sacrificing the picking stability.

Detection results

We evaluate the detection performance of PALM by comparing with the public SCSN catalog (Hutton *et al.*, 2010; Hauksson *et al.*, 2020) and the catalogs of Ross, Idini, *et al.* (2019) and Shelly (2020).

We first compare the number of detections in the source area (117.28°–117.8° W and 35.49°–36.01° N). PAL detected 27,024 events, ~80% more than that in the SCSN catalog, showing higher detection ability (Fig. 6a). The relocated MESS catalog contains 59,159 events, a comparable number to that in Ross, Idini, *et al.* (2019) and more than 1.5 times that in Shelly (2020) (Fig. 6b). Ross, Idini, *et al.* (2019) employed templates from 4 July to 24 July 2019, which is eight days more than that used by Shelly (2020) and this study. We also measure the recall rate of PALM catalog and the catalogs of Ross, Idini, *et al.* (2019) and Shelly (2020) on the SCSN catalog to estimate the detection completeness, which results in 90.05%, 94.83%, and 93.53% recall rate for the three catalogs, respectively. A slightly higher recall rate for the catalogs of Ross, Idini, *et al.* (2019) and Shelly (2020) is reasonable because they use the SCSN catalog as a template, and the matched filter tends to recall template events themselves (i.e., self-detection). Most of the missed events come

Figure 5. (a) Phase-picking performance of the PAL, STA/LTA, and kurtosis algorithms are evaluated as deviations from the Southern California Seismic Network (SCSN) picks and plotted as histograms using slash-filled, gray-filled, and blank bars, respectively. Detection accuracy is defined as the ratio of picks with an absolute deviation of 0.5 s. Phase-picking precision is presented by the mean $\pm$ standard deviation (st. dev.; s) value for the detected phases. (b) P-wave detection accuracy and picking precision for different window lengths. Solid lines with dot, triangle, and square markers denote STA/LTA with the long window length set to 2, 4, and 6 s, respectively. Dashed lines with circle marker denote picking result by kurtosis picker. In the upper panel, blue and red lines plot phase-detection accuracy assuming a maximum picking deviation (dt) of 0.5 and 0.2 s, respectively. In the lower panel, blue and red lines plot mean and st. dev. values of picking deviation, respectively. Adopted parameters are highlighted by yellow-filled dots. Panel (c) is the same as in panel (b), but for S pick. The color version of this figure is available only in the electronic edition.

from a poor event connection in the relocation process. It is worth noting that the detection number by MFT depends on the association process as well (see the MESS Relocation section), which determines the robustness in relocation. Thus, a small difference in the number of events can be neglected in evaluating the quality of the catalog.

The frequency–magnitude distribution is compared in Figure 6 to examine the consistency along the magnitude dimension. The SCSN catalog presents a significant b -value change for events between M_L 1–3.4 and M_L 3.5–6, which violates the Gutenberg–Richter law (Gutenberg and Richter, 1944), indicating incompleteness for events smaller than M_L 3.4. This bias does not exist in the PAL catalog, which provides

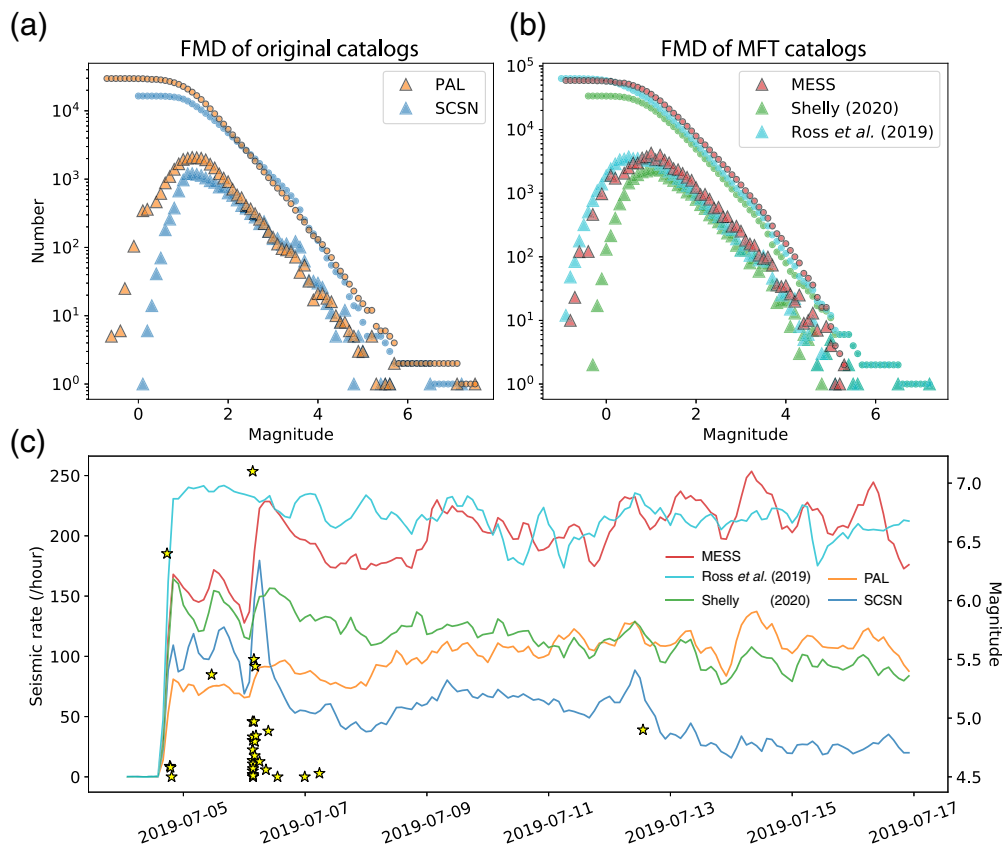


Figure 6. (a) Comparison of frequency–magnitude distribution (FMD). PAL and SCSN catalogs are plotted with yellow and blue symbols, respectively. Cumulated and noncumulative distribution are plotted as circles and triangles, respectively. Panel (b) is the same as in panel (a), but for the MFT-detected catalogs: MESS, Ross, Idini, et al. (2019), and Shelly (2020), which are plotted as red, green, and cyan symbols, respectively. (c) Seismic rate comparison. Results from different catalogs are plotted in the same colors as panels (a,b). Significant events ($M_L > 4.5$) are marked by yellow asterisks. The color version of this figure is available only in the electronic edition.

a better estimation for b -value (Wiemer and Wyss, 2000; Woessner and Wiemer, 2005). After the matched filter, the complete magnitude of the catalog of Ross, Idini, et al. (2019) is pushed to ~ 0.5 under maximum curvature criteria (Wiemer and Wyss, 2000), but it is obvious that the bias between M_L 0.5 and 2.0 still exists, which may alter further analysis on b -value. This problem does not exist in the result by MESS and Shelly (2020), although MESS shows much higher detectability. Thus, the quality of template catalog determines the matched-filter result, and the PALM architecture renders good matched-filter performance.

The temporal variation of the detection rate is compared in Figure 6c to examine the temporal consistency. Expected temporal behaviors of an early aftershock sequence include (1) abrupt increase immediately after a significant event and geometrical decrease further afterward, as described by the Omori's law (Utsu, 1961, 1970), and (2) consistent overall seismicity rate over a short period (16 days, in this study), although some possible fluctuations may be included. However, the

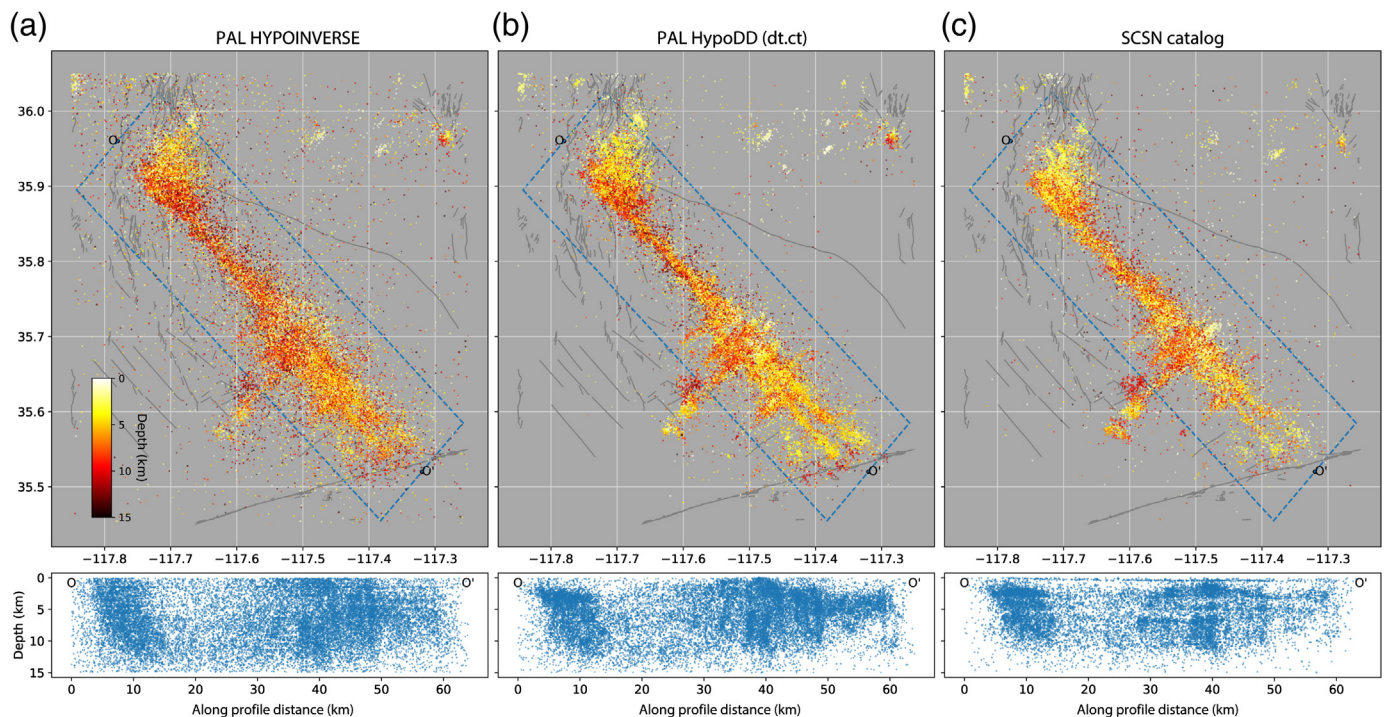
initial detections, that is, the SCSN and PAL catalogs, show systematic bias (Fig. 6c and Fig. S8): the SCSN catalog shows an artificial drop in the detection rate after 13 July 2019. This phenomenon is not reflected by other catalogs; thus, it reflects some temporal detection issue of the SCSN catalog. The PAL catalog does not show a sharp seismic activity increase after the major events, that is, the M_w 6.4 foreshock and the M_w 7.1 mainshock, indicating a magnitude completeness change during intensive seismic activities right after major events. After MFT augmentation, the detection rate in MESS shows a sudden increase after large events (Fig. 6c). This abrupt increase can also be seen in the catalog of Shelly (2020), yet it is not present in the catalog of Ross, Idini, et al. (2019) after the mainshock.

Location results

The location of the PAL and SCSN catalogs is compared in Figure 7. The PAL + HypoDD catalog qualitatively

shows a similar level of location clustering and lineation as the SCSN catalog, which is more concentrated than the HYPOINVERSE catalog (Fig. 7a), but the PAL cross section does not show concentration at certain depths as in the SCSN catalog (Fig. 7c and Fig. S10), which is a common artifact in the absolute location with 1D velocity model. The PAL + HypoDD relocation (Fig. 7b) clearly outlines basic fault structures in RC, for example, the orthogonal subfaults cutting the northwest-striking main fault and the two branches in the southeast side. In comparison with the SCSN catalog, the relocated PAL catalog recovers more detailed and complete fault geometrical structure features due to the abundance of detections. This also leads to more robust matched-filter detection throughout the study region, especially for the southeast end of the mainshock rupture, for which seismicity in the SCSN catalog is relatively sparse (Fig. 7c).

The relocated MFT catalogs, that is, the MESS catalog and the catalogs of Ross, Idini, et al. (2019) and Shelly (2020), are shown in the same manner in Figure 8. Detailed fault



structures are consistently revealed by the three MFT catalogs, including the location and strikes of secondary faults perpendicular to the main fault, the deep clusters on Garlock fault, and the complex swarm in the northwest end of the coseismic rupture. Owing to the high-resolution CC-based differential time measurement, a comparable resolution is achieved by the three MFT catalogs. The relocation resolution of the MESS catalog and the catalog of Shelly (2020) appears to be higher than that of Ross, Idini, *et al.* (2019), which may be caused by the difference in the relocation algorithms, that is, Ross, Idini, *et al.* (2019) is from GrowClust (Trugman and Shearer, 2017), but MESS and Shelly (2020) are by HypoDD (Waldhauser, 2001). The HypoDD algorithm tends to remove events showing in-consistent arrivals with other events, whereas the GrowClust algorithm treats them as low-weighted events.

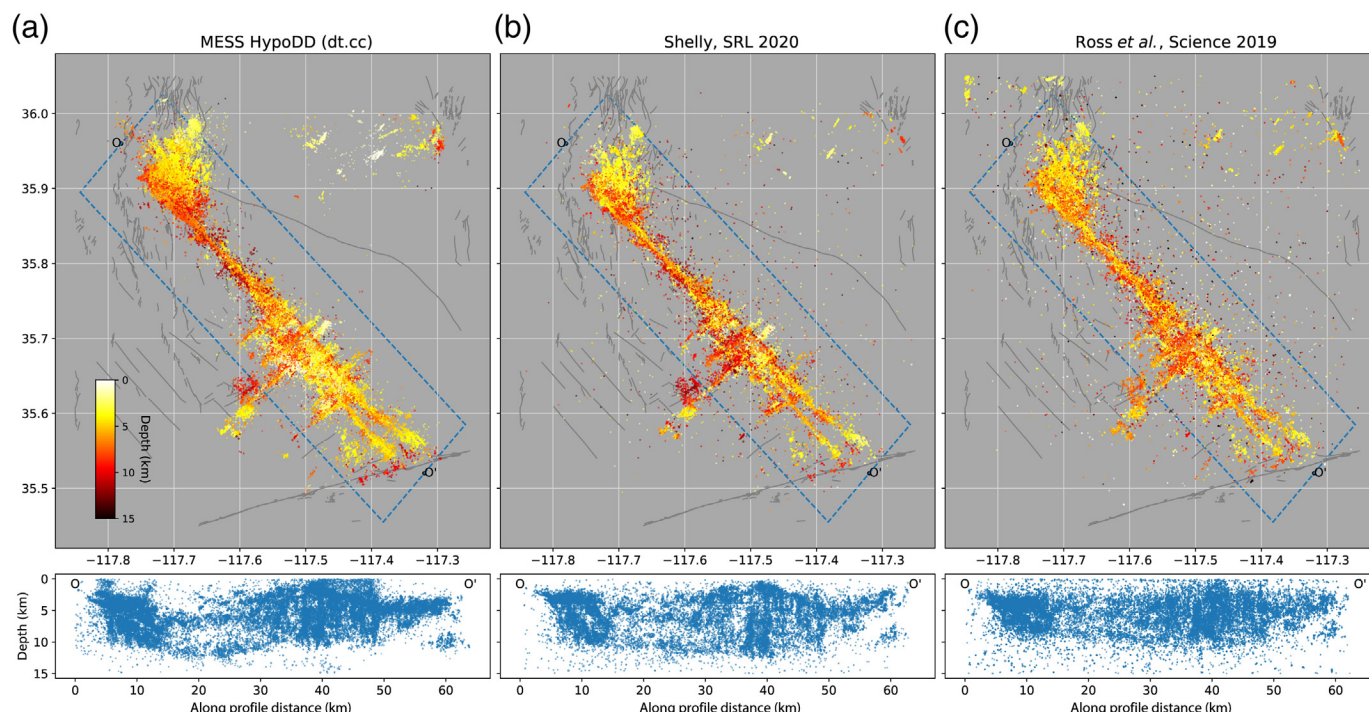
Despite the consistency of general patterns of the three catalogs, the depth distribution of these catalogs varies. The most significant difference occurs in the northwest side of the main-shock epicenter (~15–25 km along-profile distance, Fig. 8), in which large coseismic slip is observed at the shallow portion (~0–5 km, (Ross, Idini, *et al.*, 2019; Goldberg *et al.*, 2020; Jin and Fialko, 2020). In such an area, we expect to see a deficit in aftershock activity (Wetzler *et al.*, 2018), and the MESS catalog fits this pattern better than the results of Ross, Idini, *et al.* (2019) or Shelly (2020). This may come from the difference in phase association and the absolute location process, which are not included in the research of Ross, Idini, *et al.* (2019) or Shelly (2020); thus it serves as a good case to show the advantage of a complete detection and location workflow. Differences also exist in the relocation process; though both start from SCSN locations, the catalogs of Ross, Idini, *et al.* (2019) and Shelly

Figure 7. Location comparison for template events. Plot locations of catalogs obtained by (a) PAL + HYPOINVERSE, (b) PAL + HypoDD, and (c) SCSN, respectively. Earthquakes are plotted in dots color coded by their hypocentral depths with the marker size scaled with magnitude. Cross sections are plotted in blue-dashed squares. Profiles of catalog depth distribution along profile OO' are denoted as blue dots. The color version of this figure is available only in the electronic edition.

(2020) deviate from each other for ~2 km in the depth distribution (Fig. 8 and Fig. S10). Again, this is partly due to different relocation methods because GrowClust does not explicitly constrain the separation distance of event pairs. Such a deviation can be enlarged with different template locations. However, in the relocation process, we are keeping track of the consistency between MESS and the relocated templates (i.e., PAL + HypoDD), so no systematic deviations are introduced. Thus, the small differences between the MESS catalog and the catalogs of Ross, Idini, *et al.* (2019) and Shelly (2020) reflect the uncertainty introduced by the relocation method and starting locations of template events.

Conclusion

We developed an earthquake detection and location architecture that sequentially perform PALM techniques. The design of PALM is specified to directly apply to continuous seismograms and to produce high-resolution catalogs, while not requiring a reference catalog. This architecture is particularly useful for large-scale seismicity monitoring networks without detailed visual inspection, such as the on-building seismicity monitoring network in the Sichuan–Yunnan Province of China (Wu *et al.*,



2019). We adopted PALM to the 2019 Ridgecrest aftershock sequence and compared the catalog with several published catalogs. The comparison demonstrates the following:

1. The PAL picker realizes robust phase detection and high picking precision, with an optimal combination of STA/LTA and the kurtosis algorithm.
2. PAL achieves high detectability and accurate location, which serves as a sound foundation for the subsequent matched-filter detection.
3. MESS augments the PAL detections and gives high-resolution relocation with cross correlation.
4. PALM can recover an unbiased and fine fault structure from continuous data.

Data and Resources

Seismic data and phase arrivals used in this study were downloaded from Southern California Earthquake Data Center (SCEDC) at <https://scedc.caltech.edu/> (last accessed November 2020). The software phase picking, phase association, location, and matched-filter (PALM) techniques, that is, PAL and MESS, were developed with Python, Obspy, Numpy, Scipy, and Pytorch. PAL and MESS are accessible via GitHub: PAL (DOI: [10.5281/zenodo.4916097](https://doi.org/10.5281/zenodo.4916097); <https://github.com/YijianZhou/PAL>, last accessed July 2021) and MESS (DOI: [10.5281/zenodo.4916100](https://doi.org/10.5281/zenodo.4916100); <https://github.com/YijianZhou/MESS>, last accessed July 2021). The supplemental material accompanying this article includes details of the principal component analysis (PCA) polarization analysis used in the PAL picker, the efficiency in each step of PALM, and figures that help to validate the detection and location result in Ridgecrest. The detected and relocated seismic catalog by PALM can also be accessed from the supplemental material.

Figure 8. Location comparison for MFT catalogs. Markers have the same meaning as Figure 7. (a) MESS HypoDD, (b) Shelly (2020), and (c) Ross, Idini, et al. (2019). The color version of this figure is available only in the electronic edition.

Declaration of Competing Interests

The authors acknowledge that there are no conflicts of interest recorded.

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